



STEP-BY-STEP EXECUTION

● STEP 1 — Download MPJ Express

Download:

`mpj-v0_44.zip`

Extract to:

`C:\mpj`

Example final path:

`C:\mpj\mpj-v0_44`

● STEP 2 — Create Java Project

In Eclipse:

`File → New → Java Project`

Project name:

`DS_Berkeley`

● STEP 3 — Add MPJ Jar to Project

Right click project:

`Build Path → Configure Build Path`

Open:

Libraries

tab.

Click:

Add External JARs

Select:

C:\mpj\mpj-v0_44\lib\mpj.jar

Click:

Apply and Close

● STEP 4 — Create Berkeley.java

Inside src :

New → Class

Name:

Berkeley

Paste your uploaded code.

● STEP 5 — Save File

Ctrl + S

● STEP 6 — Compile from CMD

Open CMD.

Go to src folder:

```
cd C:\Users\HP\eclipse-workspace\DS_Berkeley\src
```

● STEP 7 — Compile Program

Run:

```
javac -cp ".;C:\mpj\mpj-v0_44\lib\mpj.jar" Berkeley.java
```

☒ Expected

No errors.

● STEP 8 — Set MPJ Environment

Go to MPJ bin folder:

```
cd C:\mpj\mpj-v0_44\bin
```

● STEP 9 — Run Berkeley Algorithm

Run:

```
mpjrun.bat -np 4 Berkeley
```

Meaning

Part

mpjrun.bat

-np 4

Berkeley

Meaning

MPJ runner

Run 4 processes

Main class

VERY IMPORTANT

Before running:

Copy:

```
Berkeley.class
```

from:

```
DS_Berkeley\src
```

to:

```
C:\mpj\mpj-v0_44\bin
```

OR easier method:

Run directly from src using:

```
mpjrun.bat -cp . -np 4 Berkeley
```

inside src folder.

EXPECTED OUTPUT

Example:

```
Process 0 initial clock = 10
Process 1 initial clock = 15
Process 2 initial clock = 20
Process 3 initial clock = 25
```

```
Average Clock Time = 17
```

```
Process 0 adjusted by 7 → New Clock = 17
Process 1 adjusted by 2 → New Clock = 17
Process 2 adjusted by -3 → New Clock = 17
Process 3 adjusted by -8 → New Clock = 17
```

Understanding Your Code

MPI.Init()

Starts MPI environment.

```
MPI.Init(args);
```

Rank

Each process gets unique ID.

```
MPI.COMM_WORLD.Rank();
```

Allgather()

Collects clocks from all processes.

```
MPI.COMM_WORLD.Allgather(...)
```

Master Process

Process:

```
rank == 0
```

acts as master.

Bcast()

Broadcasts average to all processes.

```
MPI.COMM_WORLD.Bcast(...)
```

COMPLETE FLOW

Start MPI

↓

Processes get clocks

↓

All clocks gathered

↓

Master computes average

↓

Average broadcasted

↓

All processes adjust clocks

↓

Synchronization complete